Northeastern University
College of Professional Studies

Northeastern University

Risk Management for Analytics – ALY 6130

Module Assignment 04: Quantitative Risk Assessment

Prepared by: Group-1

Presented to: Abeba Nigussie Turi

Date: March 16th, 2026

Overview

This report presents a quantitative risk assessment for three high-priority risks identified in the Enterprise Risk Management (ERM) framework for the Manulife–Mahindra 50:50 Life Insurance Joint Venture in India. The three risks were selected based on their Risk Score (Likelihood × Impact = 56) and strategic significance to the JV's market entry. The Indicators & Warning (I&W) analysis technique is applied to each risk, supplemented by Monte Carlo simulation (n=10,000) and a Random Forest ML classifier to quantitatively estimate probability and financial impact.

The three risks assessed are:

- R1 — LIC & Private Competitor Displacement (Competitive Risk, Score: 56)
- R9 — IRDAI Regulatory Approval Delay (Operational Risk, Score: 56)
- R15 — AI Underwriting Bias in Rural Demographics (Technology/AI Risk, Score: 56)

I&W Analysis Methodology

The Indicators & Warning (I&W) technique involves identifying observable signals — both leading and lagging — that indicate a risk is beginning to materialize. Each indicator is assigned a warning threshold. When thresholds are breached, escalation protocols are triggered. This approach transforms qualitative risk flags into quantifiable, monitorable metrics, enabling proactive risk management.

For each risk, we define: (1) key indicators drawn from industry data and internal KPIs, (2) current signal values, (3) warning thresholds that trigger a response, and (4) a warning level classification (Green / Amber / Red). Monte Carlo simulation (10,000 iterations) was then used to model probability distributions and financial exposure, while a Random Forest classifier (97% cross-validated accuracy) validated severity classifications.

Risk 1 — LIC & Private Competitor Displacement

Context

LIC commands over 62% of India's life insurance market, supported by deep government trust and a massive agent network. Private players like HDFC Life and SBI Life are also aggressively expanding into Tier-2 and Tier-3 cities — the same markets the JV targets. If the JV fails to differentiate on price, product, or distribution, it risks being displaced before gaining meaningful market share. The JV requires a minimum of 1.5% market share in Year 1 to be financially viable.

I&W Analysis Table

Indicator	Current Signal	Warning Threshold	Warning Level
LIC market share in Tier-2/3 cities	~62% (FY2024)	Market share > 70%	RED
JV new policy count (monthly)	Baseline: 0 (pre-launch)	< 5,000/month at Month 6	AMBER
Competitor premium pricing index	Industry avg ₹8,200/year	JV price > 15% above avg	AMBER
Agent recruitment rate	Target: 500/month	< 250 agents/month	RED
Customer acquisition cost (CAC)	Benchmark: ₹1,800	CAC > ₹2,500	AMBER

Monte Carlo Simulation Results (n=10,000)

Stochastic variables — competitor market share growth (Normal: μ=2.1%, σ=0.8%), JV agent ramp-up speed (Triangular: min=200, mode=500, max=900/month), and pricing premium vs market (Normal: μ=8%, σ=4%) — were simulated across 10,000 scenarios to model JV Year-1 market share outcomes.

Simulation results indicate an 84.1% probability that the JV will fall below the 1.5% viability threshold in Year 1. This is the highest materialization probability among the three risks assessed. Mean revenue at risk is USD $6.6M, rising to USD $14.1M at the P90 (worst-case 10%) level. The high probability reflects the aggressive market share target relative to LIC's entrenched dominance and the time required to build an agent network from zero.

Risk 2 — IRDAI Regulatory Approval Delay

Context

The JV requires a composite insurance license from the Insurance Regulatory and Development Authority of India (IRDAI). Regulatory delays are common in India's financial services sector — the average approval timeline for foreign-domestic JVs is 14.2 months, but complex structures involving foreign entities can take significantly longer. Every month of delay directly translates to lost revenue, deferred hiring, and ongoing holding costs, with no premium income to offset them.

I&W Analysis Table

Indicator	Current Signal	Warning Threshold	Warning Level
IRDAI application status	Submitted (Q1 2026)	No response after 6 months	AMBER
Number of regulatory queries received	0 (as of filing)	> 5 queries issued	AMBER
Foreign ownership compliance score	Fully compliant (49% cap met)	Any compliance deviation	RED
Industry avg license approval time	14.2 months (FY2024)	Exceeds 18 months	AMBER
Regulatory liaison engagement	Active (2 meetings/quarter)	No engagement > 60 days	AMBER

Monte Carlo Simulation Results (n=10,000)

Approval timeline was modeled using a log-normal distribution (μ=14.2 months, σ=3.6 months) fitted to historical IRDAI approval data for foreign-domestic JVs (2015–2024). Monthly holding cost was modeled as Normal (μ=USD $2.8M, σ=USD $0.4M). Total exposure = approval timeline × monthly cost.

Simulation results show a 17.8% probability of delays exceeding the 18-month warning threshold — the lowest probability among the three risks. However, R9 carries the highest financial exposure: mean total cost of USD $41.2M and a P90 exposure of USD $57.5M. This reflects the compounding nature of holding costs — each additional month is multiplicative. Even a moderate delay scenario generates material financial damage, making this risk financially the most consequential despite its lower probability.

Risk 3 — AI Underwriting Bias in Rural Demographics

Context

The JV plans to deploy AI-driven underwriting to process policy applications at scale — particularly in rural India where traditional documentation may be sparse or non-existent. If the AI model is trained predominantly on urban or historical data (currently < 12% rural representation), it may systematically produce biased decisions for rural applicants, creating legal liability, regulatory fines from IRDAI, and reputational damage that directly undermines the JV's core market strategy.

I&W Analysis Table

Indicator	Current Signal	Warning Threshold	Warning Level
Model training data — rural representation	< 12% rural records	Rural data < 20% of training set	RED
AI model rejection rate (rural applicants)	Benchmark < 15%	Rejection rate > 25%	RED
False positive rate on fraud detection	Industry avg: 4.2%	False positive > 8%	AMBER
Demographic parity score (fairness metric)	Current: ~0.75	Score < 0.80 (IRDAI threshold)	RED
IRDAI AI audit compliance flag	No flags	Any formal flag issued	RED

Monte Carlo Simulation Results (n=10,000)

The demographic parity score was modeled using a Beta distribution ($\alpha=6$, $\beta=2$), reflecting the current low rural data coverage. IRDAI's fairness threshold is a parity score of 0.80. Financial impact components — legal costs (LogNormal: μ=USD $8M), regulatory fine (Uniform: USD $1M–$10M), and reputational revenue loss (Normal: μ=USD $5M) — were only triggered in scenarios where the parity score fell below threshold.

Simulation results show a 57.9% probability that the AI model will breach IRDAI's fairness threshold — the second highest probability among the three risks. Mean financial impact across all scenarios is USD $11.0M, rising to USD $23.2M at the P90 level. The conditional mean impact (only in scenarios where bias is detected) is significantly higher, reflecting the compounding effect of simultaneous legal, regulatory, and reputational costs. Mitigation through rural data augmentation prior to launch is projected to reduce this probability substantially.

Quantitative Assessment Summary

The table below consolidates actual Monte Carlo simulation results (n=10,000) and ML model outputs for all three risks:

Risk	MC Probability	Mean Exposure	P90 Exposure	ML Severity	I&W Level
R1 — Competitor Displacement	84.1%	USD $6.6M	USD $14.1M	Medium	RED
R9 — Regulatory Delay	17.8%	USD $41.2M	USD $57.5M	Medium	AMBER
R15 — AI Underwriting Bias	57.9%	USD $11.0M	USD $23.2M	Medium	RED

Conclusion & Key Findings

The quantitative assessment reveals three materially different risk profiles despite identical risk scores (56) in the register:

- R1 (Competitor Displacement) is the most likely to materialize (84.1%) but carries the lowest financial exposure ($14.1M P90). The JV's Year-1 market share target is extremely aggressive against LIC's entrenched position.
- R9 (Regulatory Delay) has the lowest probability (17.8%) but by far the highest financial exposure ($57.5M P90). When it occurs, it is the most damaging risk of the three due to compounding monthly holding costs.
- R15 (AI Bias) sits in the middle — 57.9% probability and $23.2M P90 exposure. It is uniquely dangerous because it directly undermines the JV's core rural market strategy and attracts simultaneous legal, regulatory, and reputational consequences.

The ML model classified all three risks as Medium severity — consistent with the Risk Register. However, Monte Carlo simulation reveals that ML severity labels alone are insufficient for decision-making: two risks (R1 and R15) have materially high probabilities requiring immediate pre-launch mitigation, while R9 requires a dedicated financial contingency reserve despite its lower likelihood.

5. Three-Dimensional Risk Prioritization: Incorporating Control Level

5.1 Methodological Extension

The quantitative risk assessment in Section 4 establishes a strong foundation through Monte Carlo simulation and I&W analysis. However, two risks sharing an identical Risk Score do not necessarily warrant identical management urgency. Following the triangulation framework proposed by *Jalilvand and Moorthy (2023)*, this section extends the risk prioritization model by incorporating a third dimension: **Control Level**, scored using the Risk and Insurance Management Society's five-level Risk Maturity Model (RIMS RMM).

Control Level reflects the degree to which existing mitigation measures are formalized, integrated, and strategically embedded within the JV's operating model. The five levels are: **Ad hoc (1)** — improvised individual actions; **Initial (2)** — siloed, no integration; **Repeatable (3)** — established framework, operational level; **Managed (4)** — enterprise-wide monitoring harmonized; **Leadership (5)** — embedded in strategy with defined risk appetite thresholds.

Each of the 32 risks was assigned a Control Level score based on the specificity and integration of its documented mitigation strategy. Two priority indices were then computed: (1) **EI (Expected Impact)** = Likelihood × Impact; and (2) **CA (Control-Adjusted)** = EI ÷ Control Level, surfacing risks where high exposure is compounded by weak mitigation. A **Composite Priority Rank** was derived as the equally weighted average of the EI and CA ranks, following the multi-index approach in Jalilvand and Moorthy (2023).

5.2 Control Level Scoring (Selected Risks)

The table below documents the Control Level assigned to the ten most strategically significant risks, with scoring rationale based on RIMS RMM criteria:

Risk	Name	Control	RIMS Label	Scoring Rationale
R1	LIC Competitor Displacement	3	Repeatable	Bundling strategy defined; no enterprise-wide monitoring KRI established.
R3	Regulatory Tailwinds Opp.	4	Managed	30-day and 6-month milestones defined; parallel composite license filing in progress.
R6	Operational Cost Overrun	3	Repeatable	Mahindra Finance infrastructure leverage plan and phased expansion defined.
R9	IRDAI Approval Delay	4	Managed	Dedicated IRDAI liaison, pre-application consultation, complete first-submission strategy.
R15	AI Underwriting Bias	4	Managed	Mandatory bias testing gate, AI Fairness Review Board, explainability requirements.
R16	Limited Rural AI Training Data	3	Repeatable	Mahindra Finance data partnership agreed; conservative pricing loadings defined.
R17	Competitor AI Advantage	2	Initial	Data licensing and federated learning aspirational only; no timeline or budget.
R18	Global AI Standards Gap	3	Repeatable	Modular architecture and annual benchmarking defined; no board-level governance threshold.

Risk	Name	Control	RIMS Label	Scoring Rationale
R21	Aadhaar Data Privacy Breach	4	Managed	DPO appointed; data minimization; annual third-party audits; DPDPA notification plan.
R26	AI Transparency/Explainability	4	Managed	XAI frameworks from inception; plain language explanations; proactive IRDAI engagement.

5.3 Three-Dimensional Priority Ranking (Top 15)

The composite ranking below integrates EI rank and CA rank across all 32 risks. The top 15 are shown (Score color: red = HIGH ≥45, green = MEDIUM/LOW; Control color: red = weak control, blue = strong control):

Rank	Risk	Name	Category	L	I	Score	Ctrl	EI Rank	CA Rank	Composite
1	R3	Regulatory Tailwinds Opp.	Competitive	9	9	81	4	#1	#1	#1
2	R1	LIC Competitor Displacement	Competitive	7	8	56	3	#2	#2	#2
3	R18	Global AI Standards Gap	Technology/AI	7	8	56	3	#5	#3	#3
4	R6	Operational Cost Overrun	Financial	9	6	54	3	#7	#4	#4
5	R9	IRDAI Approval Delay	Operational	7	8	56	4	#3	#9	#5
6	R16	Limited Rural AI Data	Technology/AI	9	6	54	3	#8	#5	#6
7	R15	AI Underwriting Bias	Technology/AI	7	8	56	4	#4	#11	#7
8	R2	Digital Distribution Failure	Competitive	7	6	42	3	#10	#7	#8
9	R26	AI Transparency/Explainability	Legal/Compliance	7	8	56	4	#6	#12	#9
10	R10	Agent Recruitment Failure	Operational	7	6	42	3	#11	#10	#10
11	R17	Competitor AI Advantage	Technology/AI	5	6	30	2	#17	#6	#11
12	R21	Aadhaar Data Breach	Info Security	5	9	45	4	#9	#15	#12
13	R14	Mahindra Finance Integration	Operational	5	8	40	3	#12	#13	#13
14	R8	AI Efficiency Savings Lost	Financial	7	4	28	2	#20	#8	#14
15	R30	Algorithmic Fairness / Brand	Reputational	5	8	40	3	#14	#14	#15

5.4 Key Findings

Finding 1: R18 (Global AI Standards) is more urgent than its Risk Score implies

R18 shares an identical Risk Score of 56 with R1, R9, R15, and R26. Under the standard two-dimensional ranking, all five risks are treated as equivalent HIGH-priority exposures. Under the Control-Adjusted Index, however, R18 rises to **Composite Rank #3** — above R9 and R15 — because its mitigation infrastructure remains at Control Level 3 (Repeatable). No enterprise-wide monitoring system or board-level AI governance threshold has been established. As globally funded InsurTech platforms accelerate deployment in India, this absence creates a compounding vulnerability the standard Risk Score does not capture.

Finding 2: R9 (IRDAI Delay) is less immediately urgent than its Risk Score suggests

R9 carries the highest financial exposure (P90: USD $57.5M) and ranks jointly first in the two-dimensional register. The three-dimensional analysis reassigns it to **Composite Rank #5**, because the JV has already deployed comparatively mature controls — a dedicated IRDAI liaison, pre-application consultation protocol, and complete first-submission strategy — placing it at Control Level 4 (Managed). The risk remains high-priority financially, but additional management investment here has lower marginal return than weaker-controlled risks.

Finding 3: R17 (Competitor AI Advantage) is the most significant blind spot

R17 has a Risk Score of only 30, placing it in the MEDIUM tier. The three-dimensional analysis reveals this to be misleading: R17 has the lowest Control Level of any MEDIUM or HIGH risk in the register at **Control Level 2 (Initial)**. Mitigation references data licensing and federated learning aspirationally, with no timeline, budget, or KRI. Under the CA Index, R17 rises to **Composite Rank #11 — the highest-ranked MEDIUM risk in the register**. Given that HDFC Life and SBI Life each have 15–20 years of Indian policyholder training data, this gap widens every year without a structured response.

5.5 Implications for Risk Management

The three-dimensional analysis yields three concrete recommendations:

- Establish an AI Governance Committee covering R18, R15, R17, and R26 collectively, with quarterly Board reporting and defined risk appetite thresholds.
- Develop a structured response plan for R17 before JV launch: specific budget allocation, formal data-licensing timeline with IRDAI, and model performance parity target vs. top-3 private competitors.
- Maintain the R9 financial contingency reserve, but proportionally reallocate management attention toward R18, R6, and R17 where control infrastructure is weakest.

These findings demonstrate that Monte Carlo simulation and I&W analysis, while informative on the probability and financial dimensions of risk, are incomplete without an explicit assessment of control maturity. The triangulation of Likelihood, Impact, and Control Level provides a more accurate basis for management prioritization, consistent with the framework in Jalilvand and Moorthy (2023).